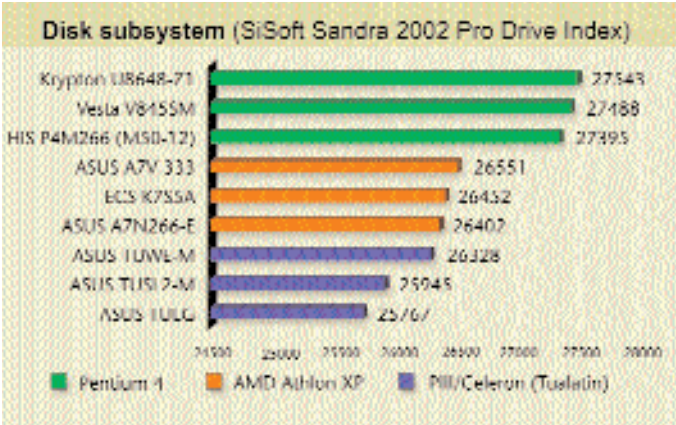




lated from the noise of other devices on the motherboard, you might experience data corruption and lock-ups when large amounts of data are suddenly pushed through the bus, as it collapses under the load.

Pentium 4: Intel's IDE controller implementation has always been excellent. This is all the more apparent in the scores of the Intel chipset-based boards which hovered above the 25000 mark in SiSoft's Drive Index. However, VIA has made good progress and has caught up with Intel, with scores around the 24500 mark. A huge surprise was when the Vesta V845SM (using the Intel 845 chipset), took the top spot by hitting a score of 27488 with a sequential read speed of around 40 MBps.

Athlon XP/Duron: The VIA based Athlon boards were quite disappointing and their scores were patchy at best. Top honours went to the ASUS A7V333—it scored a meaty 26551 in the Drive Index. **Pentium III/Celeron (Tualatin):** Strangely, the ASUS TUWE-M based on the 810E2 chipset took top marks in this category by scoring 26328.

As a general rule, any motherboard that scores over 25,000 in the SiSoft Sandra Drive Index should be considered a great performer, while the few that cross 27000 are truly exceptional.

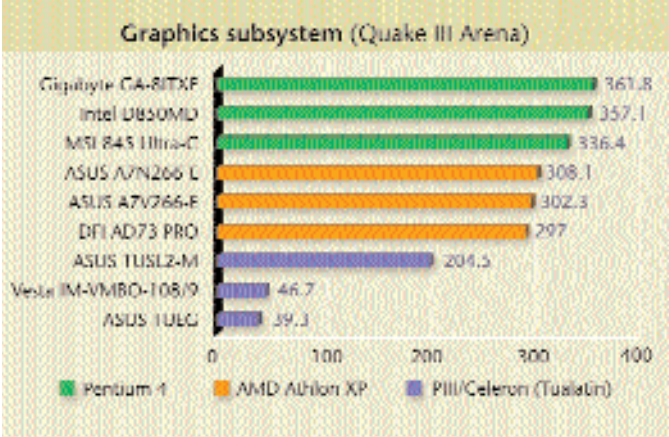
Graphics subsystem

This subsystem affects the video performance of the PC and is responsible for generating the images you see on your monitor. The bandwidth offered by the AGP slot via the AGP bus directly affects the performance of this subsystem. Also, the motherboard needs to keep the graphics card well fed with data from both the processor and the memory.

Though one can argue that the results

in these benchmarks are mostly dependant on the graphics card, you will be amazed to find large performance variances. This is due to the different chipsets and their implementation on a motherboard. **Pentium 4:** The pure bandwidth offered by the RDRAM-based motherboards has no match when it comes to gaming. The raw power of the GeForce3 is unleashed with the Gigabyte GA-8ITXE (Intel 850MD chipset) scoring a mind-bending 361.8 fps in *Quake III Arena*. The original Intel D850MD is not to be left too far behind; it scored 357.1 fps and even managed to exact some revenge in 3DMark 2001, where it scored 7476 as compared to the 7371 posted by the Gigabyte board.

The next best solution was the Intel 845D-based MSI 845 Ultra-C with DDR memory that put up a very respectable score of 336.4 fps. The gap in perform-



ance still exists, but it has been narrowed down significantly, as one can see in the 3DMark 2001 test where the MSI board actually beat the Gigabyte with a score of 7382 3DMarks.

The fact that not even a single SDRAM-based board could come close to the magical figure of 300 fps clearly shows how the graphics subsystem is affected by lack of memory bandwidth. The fastest in SDRAM board was the Intel D845WN, which trailed far behind the performance leaders with just 260.5 fps.

Athlon XP/Duron: The nForce 420-D-based ASUS A7N266-E really shines with a wonderful 308.1 fps in *Quake III Arena*. It was followed by the ASUS A7V266-E which generated 302.3 fps. No other board managed to cross the 300 fps barrier, though the DFI AD73Pro came very close at 297 fps. In the 3DMark 2001 test the A7N266-E scored a remarkable 7508 3DMarks, which outperforms even the Pentium 4 and RDRAM combo!

Pentium III/Celeron (Tualatin): Here the Pentium III platform clearly shows its age. The slow speed SDRAM doesn't allow the GeForce3 to stretch its legs and it gets heavily bottlenecked. The top score that was achieved in *Quake III Arena* was a paltry 204.5 fps by the ASUS TUSL2-M. The 3DMark score also languished at a lowly 5862 compared to the 7000+ 3DMarks achieved by several DDR and RDRAM boards in this test.

System-wide performance

This test allows you to get a clearer picture of a motherboard's impact on overall performance by testing how each subsystem interacts with each other.

Content Creation Winstone 2001 (CCWS) and **Business Winstone 2001** are

the most popular system-wide benchmarks because they touch all aspects of everyday computing. Both of them use an extensive set of applications that test the subsystems simultaneously—with the exception of the graphics subsystem which is not really stressed. Video encoding is another CPU,

memory and disk intensive benchmark that provides a good measure of your system's real world performance.

Pentium 4: From these tests, it's once again very clear that RDRAM powered boards are unmatched. The Gigabyte GA-8ITXE scores 62.1 in CCWS, winning by a hair's breadth over the Intel D850MD, which scores 62.0. It finished the video encoding sequence in a blazing 111 seconds with the Intel board taking just one second longer.

The top DDR board was the ASUS →